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EDUCATION

Ashesi University College – Berekuso, Eastern Region

BSc. Computer Science | Jan 2022 – July 2025

Ghana Christian International High School – Dodowa, Accra

General Science (WASSCE) | Aug 2018 – Oct 2021

WORK EXPERIENCE

Chartered Accountant Job Shadowing – Citibank, UK (2022)

- Gained exposure to financial analysis, budgeting, and corporate accounting.
- Observed data management practices in a professional banking environment.
- Performed data cleaning process for analysis.

S&P Global – S&P Dow Jones Indices – London, UK | August 2025

- Engaged with the Product Management team supporting global financial index products.
- Observed product lifecycle processes including research, benchmarking, and client positioning.
- Participated in discussions on market trends and index methodology.
- Gained exposure to financial data analysis and product strategy.

University of Cambridge – PACE

Data Science with Machine Learning & AI Career Accelerator

Sept 2025 – July 2026 (In Progress)

- Applied statistical modelling to solve real-world business problems.
- Implemented supervised learning techniques including regression and classification.
- Conducted exploratory data analysis and feature engineering.
- Evaluated model performance using industry-standard metrics.
- Translated analytical insights into business recommendations.

iSmart International Ghana (2025–2026)

Software Developer | Sept 2025 – Present

- Built ISmartPay, a digital wallet system with secure registration, authentication, balance management, and P2P transfers processing transactions in Ghana.
- Designed features such as dynamic service charges, transaction history, and fee collection in response to client feedback and common complaints.

- Developed backend services with Java and MySQL, optimizing database performance for balances, users, and transaction logs.
- Delivered a responsive interface ensuring usability, security, and transparency in financial workflows.

PROJECTS AND RESEARCH

Sign Language Recognition System

- Built a real-time sign language recognition system using MediaPipe Holistic and Keras.
- Combined an external dataset with self-collected gesture samples, preprocessing data through landmark extraction, coordinate normalization and noise filtering.
- Achieved 91% accuracy in recognizing 26 hand signs through model training and validation.
- Developed an interactive Streamlit app to demonstrate real-time gesture recognition.

AI Waste Detection System

- Designed and implemented a user-friendly interface AI-based waste detection system that classifies hazardous or recyclable waste with 93% accuracy.
- Trained a machine learning model to analyse uploaded images and provide eco-friendly disposal recommendations.
- Integrated the system into a Flask web app deployed on Google Cloud Platform (GCP) with Docker containerization.

Quantitative Research in Data Science – Ashesi University (Jan 2024 – May 2024)

- Designed and conducted a mixed-method survey (quantitative + qualitative) to analyze how academic course load influences student participation in extracurricular activities.
- Collected, cleaned, and pre-processed survey data from 1132 students for statistical analysis and visualization.
- Applied data visualization and statistical methods (averages, distribution plots) to uncover correlations across majors and gender.
- Generated actionable insights for university decision-making to improve student engagement.

Customer Churn Prediction System

- Developed a deep learning application (MLP, Keras Functional API) to predict telecom customer churn.
- Conducted Exploratory Data Analysis (EDA) to identify churn-driving features.
- Achieved 91% accuracy and strong AUC performance.
- Implemented hyperparameter tuning (Grid Search) to optimize model.
- Deployed the model via a Streamlit web app with real-time churn predictions.
- Packaged solution with saved model (HDF5), scaler, and encoders for reproducibility.

Predictive Modeling in Sports Analytics – Ashesi University (Jan 2024 – Apr 2024)

- Built models to predict FIFA player ratings using historical datasets.
- Performed data preprocessing, feature engineering, and cross-validation.

- Trained Random Forest, XGBoost, and Gradient Boost Regressor models.
- Evaluated performance with RMSE/MAE and tested on unseen FIFA 22 data.
- Deployed the final ensemble model on Streamlit with prediction confidence scores.

SKILLS

- Programming & Data: Python, Java, PHP, SQL, HTML/CSS, JavaScript.
- ML/AI Frameworks: TensorFlow, Keras, scikit-learn, Hugging Face, OpenCV, MediaPipe.
- Data Tools: Pandas, NumPy, Power BI.
- Cloud & DevOps: Google Cloud Platform, Docker, Git/GitHub.
- Core Strengths: Data preprocessing, ML model training, AI deployment, API integration.

CO-CURRICULAR ACTIVITIES

Ashesi University – Member of Parliament (Jan 2024 – Dec 2024)

- Implemented the 50 to 1 Initiative, ensuring each student had a designated Member of Parliament.
- Approved budgets for student councils, committees, and clubs contributing to effective financial planning.

Ashesi University – Vetting Committee (Oct 2024 – Dec 2024)

- Vetted aspirants for Student Council and Judicial Electoral Council positions.
- Assessed candidates for leadership roles for the 2025 academic year.